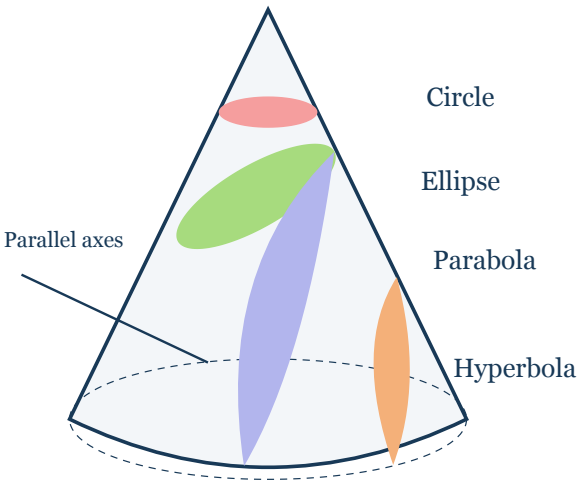
Hyperbolic Functions

Overview

We will study the definition and purpose of hyperbolic functions, such as $\sinh x$ and $\cosh x$, their inverses, and how to manipulate, differentiate and integrate them.

1. Definitions and sketches: find the exact value of $\tanh(\ln 4)$.
2. Inverse functions: prove $\operatorname{arcosh} x = \ln(x + \sqrt{x^2 - 1})$.
3. Identities and equations: solve $2\cosh^2 x - 5\sinh x = 5$ for real x.
4. Differentiation: determine $\frac{d}{dx}\operatorname{arsinh}(\sqrt{x})$.
5. Integration: determine $\int \frac{1}{\sqrt{12x+2x^2}}\,dx$.

Conic Sections 3



The axis of the parabola is parallel to the side of the cone.

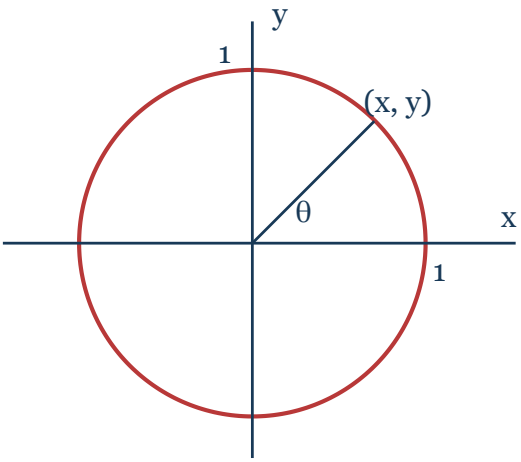
There are different families of curves. FP1 covers ellipses, parabolas and hyperbolas in Chapters 2 and 3, “Conic Sections I and II”.

Each family has different properties and equations. Slicing a cone in different ways produces these curves, hence “conic sections”.

Comparing Circles and Hyperbolas 4

Context only — no notes needed. Hyperbolas in FP1 explain the name “hyperbolic”.

The simplest circle is the unit circle centred at the origin.



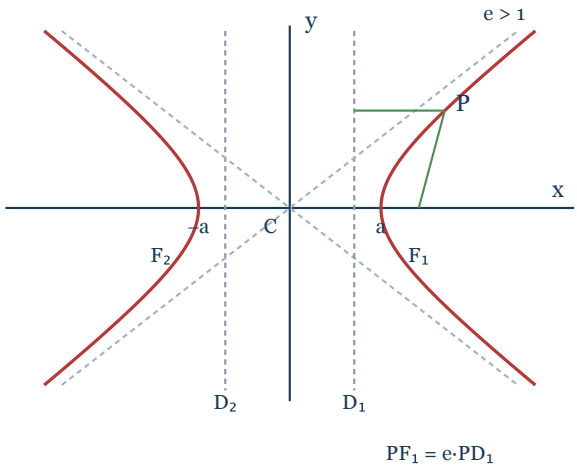
Cartesian equation:

$$x^2 + y^2 = 1.$$

Parametric equations:

$$x = \cos\theta, \quad y = \sin\theta.$$

The equivalent unit hyperbola crosses the x-axis at $(1, 0)$ and $(-1, 0)$.



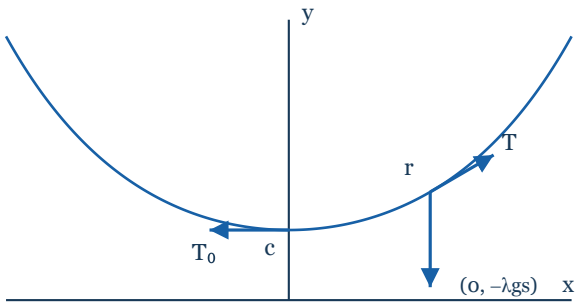
Cartesian equation:

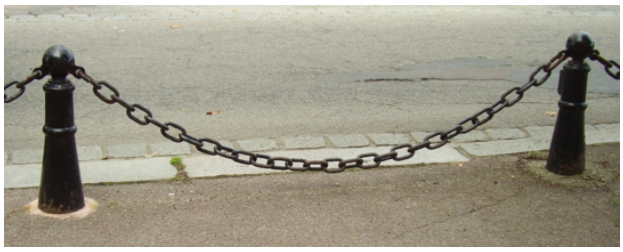
$$x^2 - y^2 = 1.$$

Similar parametric equations for the right branch:

$$x = \cosh\theta, \quad y = \sinh\theta.$$

What Is the Point of Hyperbolic Functions? 5





Hyperbolic functions often arise from differential equations, for example in mechanics. Later we will use them in calculus.

Consider the forces on each point of a hanging string: horizontal tension T_0, tangent tension T, and weight $(0, -\lambda g s)$. Solving the differential equation gives a catenary.

In normalised coordinates:

$$y = \cosh x.$$

Equations for Hyperbolic Functions 6

Hyperbolic sine ("shine"): $x \in \mathbb{R}$.

$\sinh x$

$$= \frac{e^x - e^{-x}}{2}.$$

Hyperbolic cosine ("cosh"): $x \in \mathbb{R}$.

$\cosh x$

$$= \frac{e^x + e^{-x}}{2}.$$

Hyperbolic tangent ("tanch"): $x \in \mathbb{R}$.

$\tanh x$

$$= \frac{\sinh x}{\cosh x}.$$

$$= \frac{e^{2x} - 1}{e^{2x} + 1}.$$

Hyperbolic secant ("setch"): $x \in \mathbb{R}$.

$\operatorname{sech} x$

$$= \frac{1}{\cosh x}.$$

$$= \frac{2}{e^x + e^{-x}}.$$

Hyperbolic cosecant ("cosetch"): $x \neq 0$.

$\operatorname{cosech} x$

$$= \frac{1}{\sinh x}.$$

$$= \frac{2}{e^x - e^{-x}}.$$

Hyperbolic cotangent ("coth"): $x \neq 0$.

$\coth x$

$$= \frac{1}{\tanh x}.$$

$$= \frac{e^{2x} + 1}{e^{2x} - 1}.$$

Equations for hyperbolic functions 7

Calculate $\sinh 3$ using both your calculator's hyperbolic function and the exponential formula.

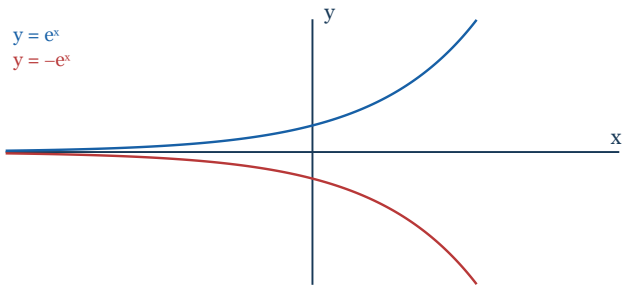
Calculator tip: use the hyp function.

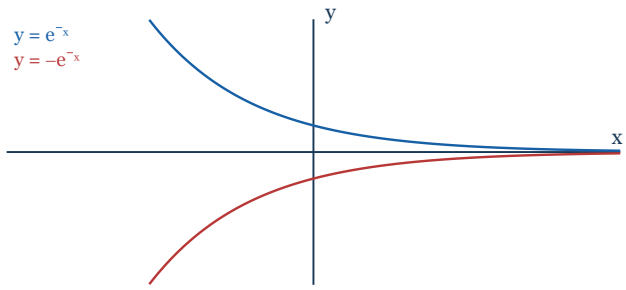
Find the exact value of $\tanh(\ln 4)$.

Write $\operatorname{cosech} 3$ in terms of e.

Solve $\sinh x = 5$.

Sketching Hyperbolic Functions 8





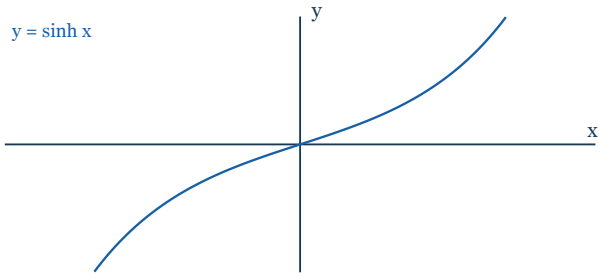
Take the average of e^x and $-e^{-x}$.

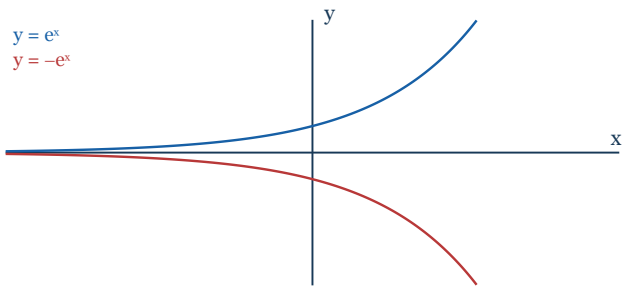
As $x \to \infty$, $e^{-x} \to 0$, so $\sinh x \sim \frac{1}{2}e^x$.

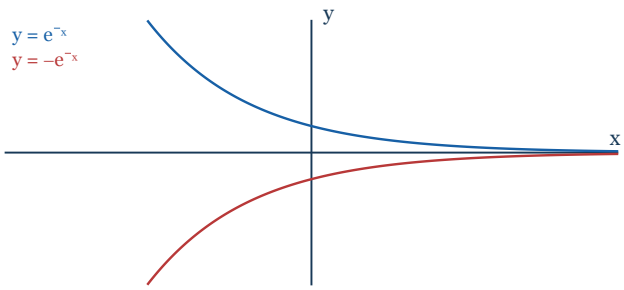
$$\sinh x = \frac{e^x - e^{-x}}{2}.$$

This is an odd function: $f(-x) = -f(x)$. Can you name others?

Other functions







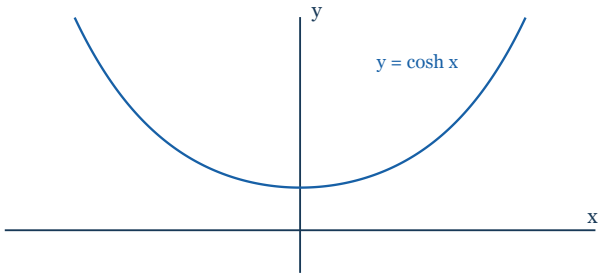
Take the average of e^x and e^{-x}.

As $x \to \infty$, $e^{-x} \to 0$, so $\cosh x \sim \frac{1}{2}e^x$.

$$\cosh x = \frac{e^x + e^{-x}}{2}.$$

This is an even function: $f(-x) = f(x)$. Can you name others?

Other functions



To sketch $y = \tanh x = \sinh x / \cosh x$, consider the usual features.

When $x = 0$:

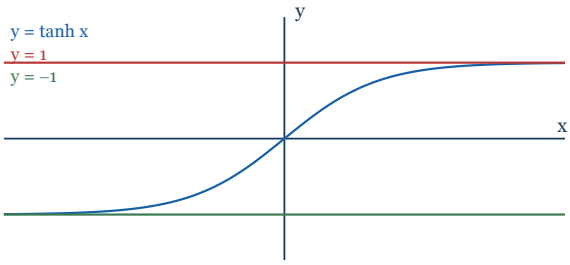
$$y = \frac{\sinh 0}{\cosh 0} = \frac{0}{1} = 0.$$

As $x \to \infty$:

$$\tanh x \sim \frac{\frac{1}{2}e^x}{\frac{1}{2}e^x} = 1.$$

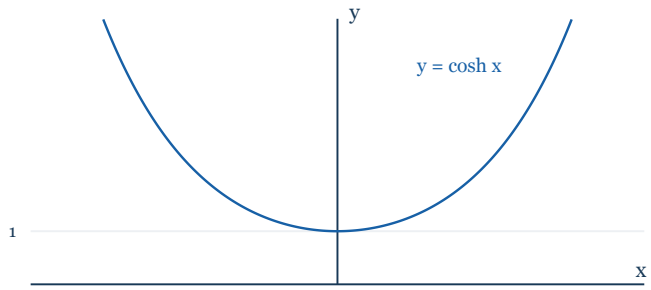
As $x \to -\infty$:

$$\tanh x \sim \frac{-\frac{1}{2}e^{-x}}{\frac{1}{2}e^{-x}} = -1.$$



Test Your Understanding 11

Sketch $y = \operatorname{sech} x$.



Reciprocal sketch

FP3 June 2011 Q5(a)

$C_1 : y = 3\sinh 2x$, $C_2 : y = 13 - 3e^{2x}$.

Sketch both on one set of axes, giving any asymptote and the points where the curves cross the axes. (4 marks)

Exam sketch

Features

Exercise 6A — Core Pure Book 2 Pages 122–123 12

Exercise 6A — Core Pure Book 2, pages 122–123.

Exercise 6A — Core Pure Book 2, pages 122–123.

Exercise 6A — Core Pure Book 2 Pages 122–123 14

Exercise 6A — Core Pure Book 2, pages 122–123.

Exercise 6A — Core Pure Book 2, pages 122–123.

Exercise 6A — Core Pure Book 2 Pages 122–123 16

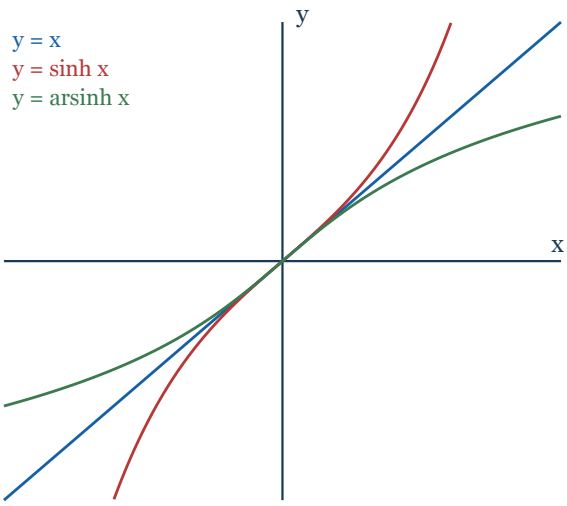
Exercise 6A — Core Pure Book 2, pages 122–123.

Each hyperbolic function has an inverse. Use the prefix "ar", as in arsinh (not arcsinh).

The six inverses are

$\operatorname{arsinh} x,\ \operatorname{arcosh} x,\ \operatorname{artanh} x,$

$\operatorname{arsech} x,\ \operatorname{arcosech} x,\ \operatorname{arcoth} x.$



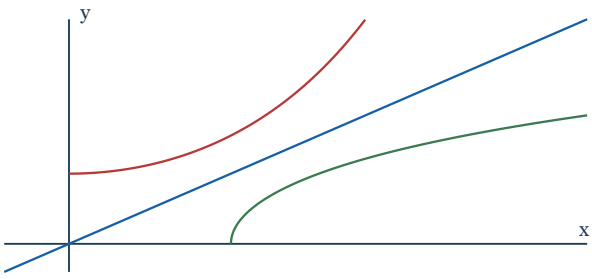
$y = \sinh x.$

Reflect in $y = x$ to get $y = \operatorname{arsinh} x$.

Why is there a problem finding the inverse of $f(x) = \cosh x$?

A function has an inverse only if it is one-to-one. Unrestricted $\cosh x$ is many-to-one, so reflecting its whole graph would give a one-to-many relation.

Restrict $x \geq 0$. Then reflect in $y = x$ to get $y = \operatorname{arcosh} x$, with domain $x \geq 1$.



y = x
y = cosh x (x ≥ 0)
y = arcosh x

Inverse Hyperbolic Functions 19

Hyperbolic functions are exponential, so their inverses can be expressed using logarithms.

Prove $\operatorname{arsinh} x = \ln(x + \sqrt{x^2 + 1})$.

Quadratic

Root

Logarithm

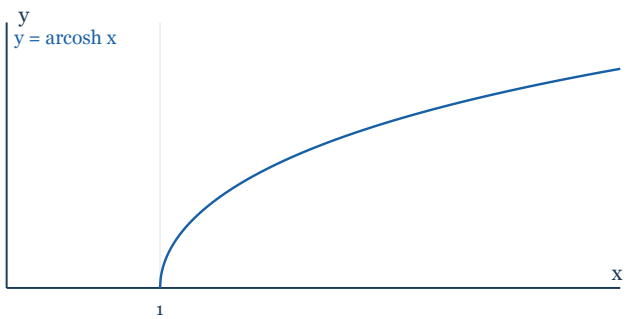
Test Your Understanding 20

Prove $\operatorname{arcosh} x = \ln(x + \sqrt{x^2 - 1})$, $x \geq 1$.

Proof

Show why the other logarithm is non-positive.

Other root

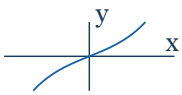


Summary So Far

$\operatorname{arsinh} x = \ln(x + \sqrt{x^2+1})$ $\quad \operatorname{arcosh} x = \ln(x + \sqrt{x^2-1}),\ x \geq 1$ $\quad \operatorname{artanh} x = \frac{1}{2}\ln\frac{1+x}{1-x},\ |x| < 1$

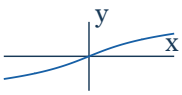
$y = \sinh x$

$x \in \mathbb{R}$



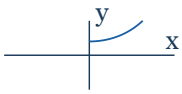
$y = \operatorname{arsinh} x$

$x \in \mathbb{R}$



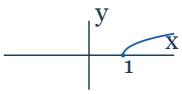
$y = \cosh x$

$x \geq 0$



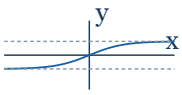
$y = \operatorname{arcosh} x$

$x \geq 1$



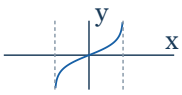
$y = \tanh x$

$x \in \mathbb{R}$



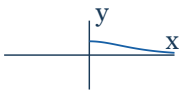
$y = \operatorname{artanh} x$

$|x| < 1$



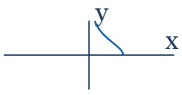
$y = \operatorname{sech} x$

$x \geq 0$



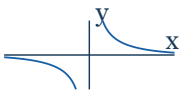
$y = \operatorname{arsech} x$

$0 < x \leq 1$



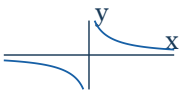
$y = \operatorname{cosech} x$

$x \neq 0$



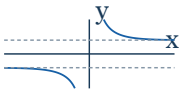
$y = \operatorname{arcosech} x$

$x \neq 0$



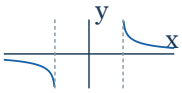
$y = \coth x$

$x \neq 0$



$y = \operatorname{arcoth} x$

$|x| > 1$



Exercise 6B — Core Pure Book 2 Pages 125

Exercise 6B — Core Pure Book 2, pages 125.

Exercise 6B — Core Pure Book 2, pages 125.

Exercise 6B — Core Pure Book 2 Pages 125 24

Exercise 6B — Core Pure Book 2, pages 125.

Exercise 6B — Core Pure Book 2, pages 125.

Exercise 6B — Core Pure Book 2 Pages 125 26

Exercise 6B — Core Pure Book 2, pages 125.

From Pure Year 1: $\sin^2 x + \cos^2 x = 1$. Are there similar hyperbolic identities?

Use the exponential definitions of sinh and cosh to prove the identities.

$\cosh^2 x - \sinh^2 x =$

1.

$\operatorname{sech}^2 x$

$= 1 - \tanh^2 x.$

$\operatorname{cosech}^2 x$

$= \coth^2 x - 1 \quad (x \neq 0).$

We can similarly prove:

$\sinh(A \pm B) = \sinh A \cosh B \pm \cosh A \sinh B.$

This resembles the sine addition formula.

$\cosh(A \pm B) = \cosh A \cosh B \pm \sinh A \sinh B.$

Here the sign is $\pm$, unlike the $\mp$ in $\cos(A \pm B)$.

Prove that

$$\tanh(A \pm B) = \frac{\sinh(A \pm B)}{\cosh(A \pm B)}$$
$$= \frac{\tanh A \pm \tanh B}{1 \pm \tanh A \tanh B}.$$

For addition the denominator has a plus sign, unlike the ordinary tangent addition formula.

Osborn's Rule 29

Osborn's rule converts trigonometric identities:

1. Replace sin by sinh, and cos by cosh.
2. Negate any explicit or implied product of two sines.

$\sin A \sin B \longrightarrow$

$-\sinh A \sinh B.$

$\tan^2 A \longrightarrow$

$-\tanh^2 A$

Since $\tan^2 A = \sin^2 A / \cos^2 A$.

$\cos 2A = 2\cos^2 A - 1 \longrightarrow$

$\cosh 2A = 2\cosh^2 A - 1.$

$$\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

$\longrightarrow$

$$\tanh(A - B) = \frac{\tanh A - \tanh B}{1 - \tanh A \tanh B}.$$

Solving Equations 30

Use identities or exponential definitions. Solve for real x:

$6\sinh x - 2\cosh x = 7.$

Solve for real x:

$2\cosh^2 x - 5\sinh x = 5.$

Further Examples 31

Pure Year 1 recap: if $\cos x = \frac{3}{5}$ and x is acute, find $\sin x$.

Recap

If $\sinh x = \frac{3}{4}$, find exactly:

(a) $\cosh x$; (b) $\tanh x$; (c) $\sinh 2x$.

a

b

c

Test Your Understanding 32

FP3 June 2009 Q1

Solve $7\operatorname{sech} x - \tanh x = 5$, giving answers as $\ln a$, where a is rational. (5 marks)

1

FP3 June 2014 (R) Q3

(a) Use exponential definitions to prove $\cosh^2 x - \sinh^2 x = 1$. (2 marks)

(b) Solve $2\sinh x + 7\cosh x = 9$ algebraically, giving natural logarithms. (5 marks)

2a

2b

FP3 June 2011 Q5(b)

Solve $3\sinh 2x = 13 - 3e^{2x}$, giving the answer as $\frac{1}{2}\ln k$, where k is an integer. (5 marks)

Solution

Exercise 6C — Core Pure Book 2, pages 128–129.

Exercise 6C — Core Pure Book 2, pages 128–129.

Exercise 6C — Core Pure Book 2 Pages 128–129 36

Exercise 6C — Core Pure Book 2, pages 128–129.

Exercise 6C — Core Pure Book 2 Pages 128–129 37

Exercise 6C — Core Pure Book 2, pages 128–129.

Differentiating Hyperbolic Functions 38

$\frac{d}{dx}\sinh x = \cosh x$

$\frac{d}{dx}\cosh x = \sinh x$

$\frac{d}{dx}\tanh x = \operatorname{sech}^2 x$

$\frac{d}{dx}\coth x = -\operatorname{cosech}^2 x.$

Memory tip: compare ordinary trig derivatives. The derivative of cosh has no minus; $(\operatorname{sech} x)' = -\operatorname{sech} x \tanh x$.

Prove $\frac{d}{dx}\sinh x = \cosh x$.

Proof

Test Your Understanding 39

June 2014 (R) Q3

The curve has equation $y = \frac{1}{2}\ln(\coth x)$, $x > 0$.

Show that $dy/dx = -\operatorname{cosech} 2x$. (3 marks)

Hint: chain rule.

Chain rule

Alternative

Inverse Hyperbolic Functions 40

$$\frac{d}{dx}\operatorname{arsinh} x = \frac{1}{\sqrt{x^2+1}}$$

$$\frac{d}{dx}\operatorname{arcosh} x = \frac{1}{\sqrt{x^2-1}} \quad (x > 1)$$

$$\frac{d}{dx}\operatorname{artanh} x = \frac{1}{1-x^2} \quad (|x| < 1).$$

Proof for arsinh:

$y = \operatorname{arsinh} x$

$\sinh y = x.$

$\frac{dx}{dy} = \cosh y.$

$\frac{dx}{dy} = \sqrt{\sinh^2 y + 1} = \sqrt{x^2+1}.$

$\frac{dy}{dx} = \frac{1}{\sqrt{x^2+1}}.$

Find $\frac{d}{dx}\operatorname{artanh}(3x)$. Use $|x| < \frac{1}{3}$.

Example 1

Given $y = (\operatorname{arcosh} x)^2$, prove $(x^2-1)(dy/dx)^2 = 4y$, for $x > 1$.

Example 2

Test Your Understanding 41

June 2009 Q4

Given $y = \operatorname{arsinh}(\sqrt{x})$, $x > 0$, find dy/dx as a simplified fraction. (3 marks)

June 2010 Q5

Given $y = (\operatorname{arcosh}(3x))^2$, $3x > 1$, show:

(a) $(9x^2 - 1)(dy/dx)^2 = 36y$. (5 marks)

(b) $(9x^2 - 1)d^2y/dx^2 + 9x\,dy/dx = 18$. (4 marks)

2a

2b

Using Maclaurin Expansions for Approximations 42

Textbook

(a) Show $\frac{d}{dx} \operatorname{arsinh} x = (1 + x^2)^{-1/2}$. We did this earlier.

(b) Find the first two non-zero terms of the Maclaurin expansion.

Given

$$\operatorname{arsinh} x = \sum_{n=0}^{\infty} \frac{(-1)^n (2n)!}{2^{2n}(n!)^2} \frac{x^{2n+1}}{2n+1}, \quad |x| < 1,$$

(c) Find the coefficient of x^5.

(d) Use terms up to x^5 to approximate $\operatorname{arsinh} 0.5$.

(e) Calculate the percentage error.

b

c

d

e

Exercise 6D — Core Pure Book 2, pages 133–134.

Exercise 6D — Core Pure Book 2 Pages 133–134 44

Exercise 6D — Core Pure Book 2, pages 133–134.

Exercise 6D — Core Pure Book 2, pages 133–134.

Exercise 6D — Core Pure Book 2 Pages 133–134 46

Exercise 6D — Core Pure Book 2, pages 133–134.

Exercise 6D — Core Pure Book 2 Pages 133–134 47

Exercise 6D — Core Pure Book 2, pages 133–134.

Standard Integrals 48

Compare the signs with ordinary trigonometric integrals. The cosech/coth results are additional context; the sech·tanh and cosech·coth results are not in the formula booklet. The arcsin and arctan results were covered in Chapter 3.

$$\int \sinh x \, dx = \cosh x + C. \qquad \int \operatorname{cosech} x \coth x \, dx = -\operatorname{cosech} x + C.$$

$$\int \cosh x \, dx = \sinh x + C. \qquad \int \frac{1}{\sqrt{1-x^2}} \, dx = \arcsin x + C, \quad |x| < 1.$$

$$\int \operatorname{sech}^2 x \, dx = \tanh x + C. \qquad \int \frac{1}{1+x^2} \, dx = \arctan x + C.$$

$$\int \operatorname{cosech}^2 x \, dx = -\coth x + C. \qquad \int \frac{1}{\sqrt{1+x^2}} \, dx = \operatorname{arsinh} x + C.$$

$$\int \operatorname{sech} x \tanh x \, dx = -\operatorname{sech} x + C. \qquad \int \frac{1}{\sqrt{x^2-1}} \, dx = \operatorname{arcosh} x + C, \quad x > 1.$$

Quickfire Examples — Do From Memory! 49

$\int \cosh(4x-1)\,dx$

$\int \sinh(\frac{2}{3}x)\,dx$

$\int \frac{3}{\sqrt{1+x^2}}\,dx$

$\int \frac{4}{\sqrt{x^2-1}}\,dx$

$\int \sinh 3x\,dx$

$\int \frac{10}{\sqrt{x^2-1}}\,dx$

$\int \frac{2}{\sqrt{1+x^2}}\,dx$

Recall the reverse chain rule, $a \neq 0$:

$$\int f'(ax+b)\,dx = \frac{1}{a}f(ax+b)+C.$$

For example,

$$\int e^{3x+2}\,dx = \tfrac{1}{3}e^{3x+2}+C.$$

For the arcosh examples use $x > 1$.

$$\int \sinh x\,dx = \cosh x + C$$
$$\int \cosh x\,dx = \sinh x + C$$
$$\int \frac{dx}{\sqrt{1-x^2}} = \arcsin x + C \quad (|x|<1)$$
$$\int \frac{dx}{1+x^2} = \arctan x + C$$
$$\int \frac{dx}{\sqrt{1+x^2}} = \operatorname{arsinh} x + C$$
$$\int \frac{dx}{\sqrt{x^2-1}} = \operatorname{arcosh} x + C \quad (x>1).$$

Further Example 50

Textbook: $\int \frac{2+5x}{\sqrt{x^2+1}}\,dx$.

Working

Integration strategy: if the numerator has multiple terms, split the fraction.

$$\int \sinh x\,dx = \cosh x + C$$
$$\int \cosh x\,dx = \sinh x + C$$
$$\int \frac{dx}{\sqrt{1-x^2}} = \arcsin x + C \quad (|x|<1)$$
$$\int \frac{dx}{1+x^2} = \arctan x + C$$
$$\int \frac{dx}{\sqrt{1+x^2}} = \operatorname{arsinh} x + C$$
$$\int \frac{dx}{\sqrt{x^2-1}} = \operatorname{arcosh} x + C \quad (x>1).$$

Integrating When Not Quite So Standard 51

Find $\int \cosh^5(2x)\sinh(2x)\,dx$.

Recall the "consider and scale" method used for $\int \sin^n x \cos x\,dx$ and $\int \cos^n x \sin x\,dx$.

Find $\int \tanh x\,dx$.

As for ordinary tan, first express the function as a quotient.

Using Identities 52

Find $\int \cosh^2(3x)\,dx$.

Use the hyperbolic double-angle formula.

Find $\int \sinh^3 x\,dx$.

This method works for small odd powers of sinh or cosh.

When That Does Not Work... 53

When a familiar trigonometric technique is unhelpful, try replacing hyperbolic functions with their exponential definitions.

Find $\int e^{2x} \sinh x \, dx$.

Find $\int \operatorname{sech} x \, dx$.

Hyperbolic Substitutions 54

For $a > 0$, choose sensible substitutions.

Recall $\sin^2\theta + \cos^2\theta = 1$, $1 + \tan^2\theta = \sec^2\theta$, and $1 + \sinh^2 u = \cosh^2 u$.

$\int \frac{dx}{\sqrt{a^2+x^2}}$

Use $x = a\sinh u$. The denominator becomes $a\cosh u$ and $dx = a\cosh u \, du$, so they cancel.

With $x = a\tan\theta$, the denominator gives $a\sec\theta$ but $dx = a\sec^2\theta \, d\theta$, leaving an extra sec factor.

$\int \frac{dx}{\sqrt{x^2-a^2}}$, $x > a$

Use $x = a\cosh u$, $u > 0$.

Show $\int \frac{dx}{\sqrt{x^2-a^2}} = \operatorname{arcosh}(x/a) + C$, $x > a$.

Proof

$$\int \frac{dx}{\sqrt{a^2+x^2}} = \operatorname{arsinh}(x/a) + C$$

$$\int \frac{dx}{\sqrt{x^2-a^2}} = \operatorname{arcosh}(x/a) + C, \; x > a.$$

Hyperbolic Substitutions 55

Textbook: show

$$\int_5^8 \frac{dx}{\sqrt{x^2-16}} = \ln\left(\frac{2+\sqrt{3}}{2}\right).$$

Working

For $a > 0$:

$$\int \frac{dx}{\sqrt{a^2+x^2}} = \operatorname{arsinh}(x/a) + C$$

$$\int \frac{dx}{\sqrt{x^2-a^2}} = \operatorname{arcosh}(x/a) + C,\ x > a$$

$$\operatorname{arsinh} x = \ln(x + \sqrt{x^2+1})$$

$$\operatorname{arcosh} x = \ln(x + \sqrt{x^2-1}),\ x \geq 1.$$

Harder Example 56

Show $\int \sqrt{1+x^2}\,dx = \frac{1}{2}\operatorname{arsinh} x + \frac{1}{2}x\sqrt{1+x^2} + C$.

Hint: use a sensible substitution. Recall $\sin^2\theta + \cos^2\theta = 1$, $1 + \tan^2\theta = \sec^2\theta$, $1 + \sinh^2 u = \cosh^2 u$.

Trigonometric substitution

Hyperbolic substitution

Test Your Understanding So Far 57

June 2013 Q2

(a) Find $\int \frac{dx}{\sqrt{4x^2+9}}$. (2 marks)

(b) Hence evaluate $\int_{-3}^{3} \frac{dx}{\sqrt{4x^2+9}}$, in the form $k\ln(a+b\sqrt{5})$, with integers a, b and constant k. (3 marks)

Hint: factor out $1/\sqrt{4}$, as in Chapter 3.

a

b

Use a hyperbolic substitution to evaluate

$$\int_0^6 \frac{x^3}{\sqrt{x^2+9}}\,dx.$$

Further example

Integrating by Completing the Square 58

Determine $\int \frac{dx}{x^2-8x+8}$.

Complete the square, then use a standard result. A substitution makes the form clearer.

Working

Standard results (formula booklet):

For $a > 0$:

$$\int \frac{dx}{\sqrt{a^2-x^2}} = \arcsin(x/a) + C, \quad |x| < a$$

$$\int \frac{dx}{a^2+x^2} = \frac{1}{a}\arctan(x/a) + C$$

$$\int \frac{dx}{\sqrt{a^2+x^2}} = \operatorname{arsinh}(x/a) + C$$

$$\int \frac{dx}{\sqrt{x^2-a^2}} = \operatorname{arcosh}(x/a) + C, \quad x > a$$

$$\int \frac{dx}{a^2-x^2} = \frac{1}{2a}\ln\left|\frac{a+x}{a-x}\right| + C$$

$$\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + C.$$

Further Example 59

Determine $\int \frac{dx}{\sqrt{12x+2x^2}}$ for $x > 0$.

Working

Standard results (formula booklet):

For $a > 0$:

$$\int \frac{dx}{\sqrt{a^2 - x^2}} = \arcsin(x/a) + C, \quad |x| < a$$

$$\int \frac{dx}{a^2 + x^2} = \frac{1}{a}\arctan(x/a) + C$$

$$\int \frac{dx}{\sqrt{a^2 + x^2}} = \operatorname{arsinh}(x/a) + C$$

$$\int \frac{dx}{\sqrt{x^2 - a^2}} = \operatorname{arcosh}(x/a) + C, \quad x > a$$

$$\int \frac{dx}{a^2 - x^2} = \frac{1}{2a}\ln\left|\frac{a+x}{a-x}\right| + C$$

$$\int \frac{dx}{x^2 - a^2} = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + C.$$

Test Your Understanding 60

June 2014 (R) Q2

$9x^2 + 6x + 5 = a(x+b)^2 + c.$

(a) Find a, b, c. (3 marks)

Hence, or otherwise, find:

(b) $\int \frac{dx}{9x^2+6x+5}$. (2 marks)

(c) $\int \frac{dx}{\sqrt{9x^2+6x+5}}$. (2 marks)

a

b

c

Exercise 6E — Core Pure Book 2, pages 140–142.

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Exercise 6E — Core Pure Book 2, pages 140–142.

Exercise 6E — Core Pure Book 2, pages 140–142.

Exercise 6E — Core Pure Book 2 Pages 140–142 64

Exercise 6E — Core Pure Book 2, pages 140–142.

Exercise 6E — Core Pure Book 2, pages 140–142.